

EEG–Hemodynamic Coupling During General Anaesthesia: A Large-Scale Exploratory Analysis in 2,070 Surgical Cases with Null Acute Event-Response Findings

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ABSTRACT

Background. Whether dynamic coupling between cortical EEG and arterial blood pressure (ABP) varies systematically across patients and responds to acute haemodynamic excursions during general anaesthesia has not been characterised at large scale. We report an exploratory retrospective analysis with predominantly null findings.

Methods. We analysed 2,070 general anaesthesia cases from VitalDB with simultaneous raw EEG (128 Hz) and invasive ABP (500 Hz). Granger causality (GC) and cross-wavelet coherence (WC) were computed in 10-minute sliding windows. A case-level composite summary score (the Neuro-Cardiovascular Coupling Index, NCCI) was derived by global PCA. A per-case Window Coupling Score (WCS), sign-aligned and MinMax-normalised within each patient, was used for event analysis. Surrogate haemodynamic excursions were defined as MAP rise >20 mmHg within 90 s with BIS <60. The primary statistical analysis was a generalised estimating equation (GEE) on per-event Δ WCS (exchangeable; sensitivity: AR(1)) to account for within-case clustering.

Results. EEG changes temporally preceded MAP changes in 71% of cases. Mean NCCI was 0.727 ± 0.067 . Surrogate excursions were detected in 2,031/2,070 cases (98.1%), indicating non-specific endpoint capture. Primary result (GEE-Exchangeable): $\beta = +0.0038$ (95% CI +0.0001 to +0.0076; $p = 0.047$), Cohen's $d = 0.047$ (negligible). With the more conservative AR(1) correlation structure $\beta = +0.0020$, $p = 0.230$ (not significant). Stricter excursion thresholds yielded null or negative effects across all definitions.

Conclusions. There is no robust evidence of WCS change at surrogate haemodynamic excursions. The predominantly null result reflects three compounding limitations: a non-specific surrogate endpoint (98.1% case rate), a mismatch between 10-minute coupling windows and 90-second excursions, and pharmacodynamic confounding uncontrolled in the current data. These findings are hypothesis-generating and highlight the methodological requirements for future coupling research with proper event labels and drug-infusion covariates.

Keywords: EEG–hemodynamic coupling, Granger causality, cross-wavelet coherence, general anaesthesia, intraoperative monitoring, null result, VitalDB, exploratory analysis

INTRODUCTION

During general anaesthesia, the brain and cardiovascular system undergo coupled physiological changes driven by anaesthetic agents (Brown et al., 2010; Purdon et al., 2015). EEG monitoring via the Bispectral Index (BIS) is widely used to track anaesthetic depth (Avidan et al., 2008), while arterial blood pressure (ABP) provides continuous haemodynamic surveillance. Whether changes in EEG activity temporally precede or follow haemodynamic changes — and whether their coupling strength varies meaningfully across patients — are clinically relevant empirical questions that have received limited systematic attention at scale.

Granger causality (GC) (Granger, 1969) is a widely applied method for testing temporal precedence between physiological time series (Pereda et al., 2005; Seth et al., 2015). Cross-wavelet coherence (WC) (Torrence and Compo, 1998) provides frequency-resolved coupling without requiring stationarity, and has been used in cardiovascular–neurological coupling studies (Pikovsky et al., 2001). The Alpha/Delta Ratio (ADR), derived from frontal EEG power spectra, provides a continuous index of cortical arousal that is sensitive to anaesthetic depth (Akeju et al., 2014; Bruhn et al., 2000).

The VitalDB open dataset (Lee et al., 2022) provides 6,388 surgical cases with synchronised EEG and ABP waveforms, offering a unique opportunity to characterise EEG–hemodynamic coupling systematically. Prior studies of intraoperative coupling have been limited to small samples or lacked the multivariate signal processing infrastructure available here.

Motivated by this gap, we conducted a planned exploratory retrospective analysis with three objectives: (1) to quantify the direction and magnitude of EEG–MAP temporal precedence (GC) and spectral coherence (WC) across a large surgical cohort; (2) to construct a case-level composite coupling summary score (NCCI) and characterise its population distribution; and (3) to test whether a per-case window-level coupling score (WCS) is associated with surrogate haemodynamic excursions, defined by MAP and BIS criteria. **This study makes no clinical validation claims and is explicitly hypothesis-generating.** The study is reported in accordance with the STROBE guideline for observational studies; the completed checklist is provided as Supplementary material.

MATERIALS & METHODS

Study Design and Data Source

Retrospective observational cohort study using VitalDB v1.0.0 (Lee et al., 2022), a high-resolution perioperative dataset of patients who underwent surgery at Seoul National University Hospital (Seoul, Republic of Korea) between August 2016 and June 2017. The original data collection was approved by the Seoul National University Hospital Institutional Review Board (H-1408-101-605), which waived the requirement for informed consent owing to the use of fully anonymised records. The dataset is openly available on PhysioNet under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence (version DOI: 10.13026/czw8-9p62); the present secondary analysis of anonymised public data required no additional ethical approval.

Patient Selection

Inclusion criteria, applied to all 6,388 cases in VitalDB v1.0.0, were: age ≥ 18 years; general anaesthesia; operation duration ≥ 60 min; and the presence of both raw EEG (128 Hz) and invasive ABP (500 Hz) waveforms with ≥ 30 min of concurrent recording. Cases from cardiac or thoracic surgical departments were excluded; no additional diagnosis-based clinical exclusions were applied. Signal-level artefact handling (EEG amplitude $> 200 \mu\text{V}$ and out-of-range ABP) is described in the next section. The selection flowchart (Figure 1) proceeds from 6,388 cases $\rightarrow$ 3,825 after the basic clinical filters (age ≥ 18 , general anaesthesia, non-cardiac/non-thoracic department, operation ≥ 60 min) $\rightarrow$ 2,076 with both EEG and ABP waveforms present $\rightarrow$ 2,070 after requiring ≥ 30 min of concurrent valid EEG and ABP signal.

Signal Processing

EEG (128 Hz): Bandpass 0.5–45 Hz (4th-order Butterworth) + 50 Hz notch. Segments $> 200 \mu\text{V}$ masked. Per 30-second window (10-second step): Welch PSD $\rightarrow \delta/\theta/\alpha/\beta$ band powers, Alpha/Delta Ratio (ADR), SEF95, spectral entropy, suppression ratio.

ABP (500 Hz): Beat-to-beat peak detection (min height 20 mmHg; min inter-peak 200 samples). Per window: MAP, SBP/DBP, pulse pressure, PPV, SDNN, RMSSD, $dP/dt_{\max}$.

ADR was selected as the primary EEG coupling variable on three grounds: (i) prior evidence that ADR tracks BIS-defined anaesthetic depth (Bruhn et al., 2000; Akeju et al., 2014); (ii) it is a continuous, interpretable scalar at each window; (iii) it showed the lowest artefact-driven outlier rate among tested features. MAP was selected as the primary haemodynamic variable because it integrates systolic and diastolic components and is the principal clinical target for haemodynamic management.

Granger Causality

Pairwise GC between ADR and MAP in 60-window (10-minute) sliding windows (3-window step) (Granger, 1969; Barnett and Seth, 2014). Stationarity: ADF test ($\alpha = 0.05$); non-stationary series first-

96 differenced. Lag: 1–5, selected by minimum SSR F-test p -value. GC_{e2h} was right-skewed (raw mean
 97 2.61, SD 5.08) and winsorised at the 99th percentile ($\hat{c} = 4.26$) for PCA input; raw values were used for
 98 directionality. GC quantifies temporal precedence, not mechanistic causation (Granger, 1969).

99 **Cross-Wavelet Coherence**

Continuous wavelet transform with a complex Morlet wavelet (cmor1.5-1.0, $dt = 10$ s) (Torrence and
 Compo, 1998). Coherence:

$$C_{xy}(a, t) = \frac{|\langle W_x W_y^* \rangle|^2}{\langle |W_x|^2 \rangle \cdot \langle |W_y|^2 \rangle} \quad (1)$$

100 Smoothed with a uniform filter ([5, 11]). Cone-of-influence (COI) edge effects affect $\approx 5\%$ of data at
 101 record boundaries; a supplementary sensitivity analysis (Supplementary S5) confirmed that removing COI-
 102 affected samples did not materially alter $WC_{overall}$ (mean change < 0.003). Summary metrics: $WC_{overall}$,
 103 WC_{low} (< 0.01 Hz), WC_{mid} (0.01–0.03 Hz).

104 **Coupling Scores: NCCI and WCS**

105 **Case-level NCCI (Neuro-Cardiovascular Coupling Index)** (one value per case): Global PCA across all
 106 2,070 cases on per-case means of [GC_{e2h} F (winsorised), $WC_{overall}$, Pearson r]. Z-score standardised; PC1
 107 MinMax to [0,1]. Sign convention: PC1 inverted if $\text{corr}(\text{PC1}, \text{WC}) < 0$ (ensuring NCCI = 1 $\equiv$ maximum
 108 WC and GC).

109 **Window Coupling Score (WCS)** (per-window; for event analysis only): Per-case local PCA on
 110 per-window [GC_{e2h} F, WC, Pearson r]. The same sign convention is applied per case. PC1 scores are
 111 MinMax-normalised within each case. WCS = 1 consistently represents higher WC and GC within that
 112 patient's own range. WCS values cannot be compared across patients and are not on the same scale as
 113 NCCI.

114 **Surrogate Haemodynamic Excursions**

115 MAP rise > 20 mmHg within 90 s while BIS < 60 . Pre-event window: the 10-minute window immediately
 116 preceding the excursion. Post-event window: the 10-minute window immediately following. This
 117 definition has known limitations: no surgical timestamps; MAP rises may reflect non-nociceptive causes;
 118 the threshold is empirically chosen. We therefore term these *surrogate haemodynamic excursions*, not
 119 nociceptive events.

120 **Statistical Analysis**

121 **Primary — GEE on ΔWCS (Exchangeable).** For each excursion, $\Delta WCS = WCS_{post} - WCS_{pre}$. A
 122 GEE with exchangeable within-case correlation (Liang and Zeger, 1986) (formula: $\Delta WCS \sim 1$, the
 123 intercept tests departure from zero), robust sandwich variance, grouped by patient ($n = 2,031$ patients,
 124 73,001 observations). This directly models the within-case relative change without requiring WCS values
 125 to be comparable across patients.

126 **Sensitivity — GEE-AR(1).** The same model with an AR(1) working correlation structure, which
 127 accounts for the temporal ordering of events within each case (more conservative for time-series data).

128 **Secondary — patient-level.** Per-patient mean ΔWCS tested against zero (one-sample t -test on patient
 129 means, $n = 2,031$). Effect size: Cohen's d .

130 **Threshold sensitivity.** The primary GEE analysis was repeated for four stricter excursion definitions
 131 (Table 2).

132 Continuous variables are reported as mean $\pm$ SD or median [IQR]. Two-sided $\alpha = 0.05$. Software:
 133 Python 3.10 (NumPy 1.24, SciPy 1.11, statsmodels 0.14, PyWavelets 1.4, scikit-learn 1.3).

134 **RESULTS**

135 **Cohort**

136 2,070 cases were analysed (Figure 1; Table 1): median anaesthesia duration 262 [195–340] min; age
 137 58.3 ± 14.5 years; 57% male; ASA I–II in 82%. A representative single-case recording, showing the
 138 synchronised EEG band powers, BIS, haemodynamics, and cross-wavelet coherence used throughout the
 139 analysis, is shown in Figure 2.

Granger Causality

Mean GC_{e2h} F (winsorised) was 2.49 ± 0.50 ; GC_{h2e} was 2.23 ± 0.50 (Supplementary Figure S1). EEG temporally preceded MAP in 71% of cases (GC directionality, winsorised: 0.28 ± 0.68). *Interpretation caveat:* given the high low-frequency WC (below), shared anaesthetic pharmacodynamic drift is a plausible primary driver of this temporal pattern; no mechanistic claim is made.

Cross-Wavelet Coherence

$WC_{overall}$ was 0.799 ± 0.034 . WC_{low} (<0.01 Hz) was 0.824 ± 0.033 , reflecting shared slow pharmacodynamic trends (periods >100 s) most likely from propofol and remifentanyl co-variation. WC_{mid} (0.01 – 0.03 Hz) was 0.352 ± 0.025 . Representative heatmaps for three NCCI tertile cases are shown in Figure 3.

Case-Level NCCI Distribution

NCCI was 0.727 ± 0.067 (median 0.729; IQR 0.694–0.766; 5th–95th pct [0.624, 0.827]). PC1 loadings: GC 0.65, WC 0.73, Pearson -0.23 ; PC1 variance 37.0%, indicating limited shared information among the three input metrics. The distribution is shown in Figure 4. Quartile stratification (Supplementary Figure S2) shows a monotonic component ordering that is a *construction artefact* of the NCCI definition and has no independent evidentiary value.

WCS Response to Surrogate Haemodynamic Excursions

Surrogate excursions were detected in 2,031/2,070 cases (98.1%), yielding 73,001 observations (median 29 per case [IQR 18–45]; max 212). The 98.1% case rate indicates this definition is non-specific and cannot discriminate nociceptive from non-nociceptive haemodynamic variation.

Primary result (GEE-Exchangeable, $n = 2,031$ patients): $\beta = +0.0038$ (SE = 0.0019, 95% CI $+0.0001$ to $+0.0076$, $z = 1.99$, $p = 0.047$). Cohen's $d = 0.047$ (negligible by conventional standards).

GEE-AR(1) sensitivity: $\beta = +0.0020$ (SE = 0.0017, 95% CI -0.0013 to $+0.0054$, $p = 0.230$). The shift from $p = 0.047$ (exchangeable) to $p = 0.230$ (AR(1)) when the temporal ordering of events is accounted for indicates that the borderline result in the primary analysis is **not robust to the correlation structure choice**.

Patient-level secondary analysis: Mean ΔWCS per patient was $+0.0047 \pm 0.0997$; pre-excursion 0.333 ± 0.124 ; post-excursion 0.337 ± 0.131 . The difference is statistically significant but of negligible magnitude and inconsistent with the primary GEE-AR(1) estimate, likely reflecting high precision from a large sample size rather than a meaningful effect ($d = 0.047$). 49% of patients showed a negative mean ΔWCS .

Cluster bootstrap: $\Delta WCS = +0.0047$; 95% CI [$+0.0004$, $+0.0090$]. The lower bound is near zero, consistent with GEE-AR(1).

Threshold sensitivity (Table 2): All four stricter definitions yield case rates $>95\%$. The $MAP \geq 25$ threshold yields $\beta \approx 0$ ($p = 0.976$); $MAP \geq 30$ thresholds yield negative β values. No threshold produced a clinically meaningful effect ($|d| \leq 0.05$). Figure 5 summarises the event analysis; the case-level Alpha/Delta Ratio analogue is shown in Supplementary Figure S3.

DISCUSSION

Principal Findings

This large exploratory study of EEG–hemodynamic coupling in 2,070 general anaesthesia cases yields one positive descriptive finding and one primary negative finding. Descriptively, EEG changes temporally precede MAP changes by GC in 71% of cases, and overall coupling coherence is high ($WC = 0.799$), predominantly at low frequencies consistent with shared pharmacodynamic co-variation. The negative finding: WCS shows no robust change at surrogate haemodynamic excursions. The GEE-Exchangeable result ($p = 0.047$) is borderline but shifts to non-significant ($p = 0.230$) with AR(1) correlation — a more appropriate structure for time-ordered events within patients. Effect sizes are negligible ($d = 0.047$) across all analyses.

The Endpoint Defines the Finding

We propose that the **primary negative result is more informative about the inadequacy of the surrogate excursion definition than about the absence of true EEG–hemodynamic coupling dynamics**. A

190 98.1% excursion rate means the definition captures nearly universal haemodynamic variation — fluid
191 administration, positional change, autonomic fluctuations — alongside any genuine nociceptive responses.
192 In such a setting, any acute coupling metric will appear insensitive, not because coupling dynamics are
193 absent, but because the endpoint cannot isolate the physiological event of interest. The threshold sensitivity
194 analysis reinforces this: stricter MAP/BIS criteria continue to capture >95% of cases, indicating the
195 problem is structural rather than threshold-dependent.

196 Window–Event Mismatch

197 A 10-minute coupling window is well-suited to characterising slow coupling trends but is fundamentally
198 mismatched to 90-second haemodynamic excursions. The WCS likely captures propofol and remifentanyl
199 pharmacodynamic co-variation at the 100–600-second timescale (Struys et al., 2000) rather than rapid
200 autonomic responses. Future work should explore coupling metrics derived from shorter windows (<2
201 minutes) or use event-locked analysis approaches rather than sliding windows.

202 GC Temporal Precedence

203 The 71% EEG-preceding result is descriptively noteworthy but difficult to interpret causally. Without
204 drug infusion covariates, the temporal structure of propofol and remifentanyl concentration trajectories
205 may introduce apparent EEG-to-MAP precedence through shared pharmacokinetics. Whether a true
206 neural coupling pathway (e.g. cortical suppression → reduced sympathetic output → MAP reduction)
207 contributes cannot be disentangled from the available data (Sessler et al., 2012).

208 Value of Negative Results in Physiological Signal Research

209 Negative and null results from large open datasets constitute scientifically valuable contributions, even
210 when they do not yield immediately actionable clinical findings. This study demonstrates: (i) that
211 EEG–MAP coupling can be systematically quantified at population scale using VitalDB; (ii) that the
212 coupling composite score NCCI captures population-level variation; and (iii) that the MAP/BIS-based
213 surrogate excursion definition is insufficiently specific for acute coupling validation in a retrospective
214 setting. This third finding — that the *endpoint* fails before the *metric* can be properly tested — is a
215 methodological contribution to future intraoperative monitoring research (Sessler et al., 2012; Struys et al.,
216 2000). Researchers designing future coupling studies should use surgical stimulus timestamps, validated
217 pain scales, or drug-intervention event logs as endpoints, rather than haemodynamic thresholds alone.

218 Limitations

219 (1) Single-centre; generalisability unknown. (2) Frontal single-channel EEG only. (3) No surgical stimulus
220 timestamps; the 98.1% surrogate excursion rate indicates a non-specific endpoint. (4) Drug infusion rates
221 not included as covariates; pharmacodynamic co-variation likely dominates the measured coupling. (5)
222 GC assumes linearity; the lag was selected by minimum p -value (which may bias toward significance;
223 fixed-lag sensitivity analyses are reported in Supplementary S3 and recommended for future work). (6)
224 NCCI and WCS are distinct constructs; WCS values are not comparable across patients. (7) 10-minute
225 WCS windows cannot track 90-second excursions. (8) No external validation cohort.

226 CONCLUSIONS

227 In 2,070 general anaesthesia cases, EEG temporally preceded MAP changes in 71% of cases and overall
228 coherence was high, predominantly reflecting shared pharmacodynamic slow trends. A case-level
229 composite coupling score (NCCI) captured systematic inter-patient variation but was self-referentially
230 constructed.

231 The **primary finding is null**: WCS shows no robust change at surrogate haemodynamic excursions
232 (GEE-Exchangeable $p = 0.047$; GEE-AR(1) $p = 0.230$; $d = 0.047$). This result reflects three compound-
233 ing limitations: endpoint non-specificity (98.1% case rate), window–event timescale mismatch, and
234 uncontrolled pharmacodynamic confounding.

235 The null finding has methodological value: it demonstrates that MAP/BIS-based surrogate excursion
236 definitions are insufficient for acute coupling validation in retrospective intraoperative data. Future
237 coupling studies should incorporate surgical timestamps, shorter coupling windows matched to event
238 timescales, and drug infusion covariates.

239 ADDITIONAL INFORMATION AND DECLARATIONS

240 Ethics

241 This study is a secondary analysis of the VitalDB v1.0.0 open dataset. The original data collection was
242 approved by the Institutional Review Board of Seoul National University Hospital (H-1408-101-605),
243 which waived the requirement for informed consent owing to the use of fully anonymised records. All
244 records are publicly available on PhysioNet under the Creative Commons Attribution 4.0 International
245 (CC BY 4.0) licence; no additional ethical approval was required for the present analysis.

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248 Competing Interests

249 The author declares that there are no competing interests.

250 Author Contributions

251 **Ge Gao** conceived and designed the study, performed the experiments, analysed the data, prepared the
252 software, and authored and approved the final manuscript.

253 Data Availability

254 VitalDB v1.0.0 is openly available on PhysioNet ([https://physionet.org/content/vitaldb/](https://physionet.org/content/vitaldb/1.0.0/)
255 [1.0.0/](https://physionet.org/content/vitaldb/1.0.0/); version DOI: 10.13026/czw8-9p62) under the Creative Commons Attribution 4.0 International
256 (CC BY 4.0) licence. No additional data were collected.

257 Code Availability

258 The full analysis pipeline (Python 3.10; NumPy, SciPy, statsmodels, PyWavelets, scikit-learn) is openly
259 available at <https://github.com/ZJUArtemis/eeg-hemodynamic-coupling>. Key scripts
260 include signal processing, GC and WC computation, NCCI/WCS construction, and statistical analysis.

261 Supplemental Information

262 Supplementary text S1–S6 and Supplementary Figures S1–S3 are provided as separate files.

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Table 1. Cohort characteristics, coupling metrics, and WCS event analysis ($n = 2,070$).

Variable	Value
<i>Demographics and clinical</i>	
Age (mean $\pm$ SD, years)	58.3 $\pm$ 14.5
Male sex	1,180 (57.0%)
BMI (mean $\pm$ SD, kg/m ²)	23.0 $\pm$ 3.7
ASA I / II / III / IV	458 / 1,201 / 342 / 24
Anaesthesia duration (median [IQR], min)	262 [195–340]
<i>Granger causality (mean $\pm$ SD)</i>	
GC _{e2h} F (winsorised)	2.49 $\pm$ 0.50
GC _{e2h} F (raw)	2.61 $\pm$ 5.08
GC _{h2e} F	2.23 $\pm$ 0.50
EEG preceding MAP	71% of cases
Pearson r (ADR vs MAP)	0.036 $\pm$ 0.181
<i>Cross-wavelet coherence (mean $\pm$ SD)</i>	
W _{Coverall}	0.799 $\pm$ 0.034
W _{Clow} (<0.01 Hz)	0.824 $\pm$ 0.033
W _{Cmid} (0.01–0.03 Hz)	0.352 $\pm$ 0.025
<i>NCCI (case-level composite summary score)</i>	
Mean $\pm$ SD	0.727 $\pm$ 0.067
Median [IQR]	0.729 [0.694–0.766]
5th–95th pct	[0.624, 0.827]
PCA PC1 variance	37.0%
PC1 loadings	GC 0.65, WC 0.73, Pearson -0.23
<i>WCS excursion analysis (per-case local PCA)[†]</i>	
Cases with ≥ 1 excursion	2,031/2,070 (98.1%)
Total excursions (observations)	73,001
Excursions/case: median [IQR]	29 [18–45]
Pre-excursion WCS (patient mean $\pm$ SD)	0.333 $\pm$ 0.124
Post-excursion WCS (patient mean $\pm$ SD)	0.337 $\pm$ 0.131
Patient-mean Δ WCS $\pm$ SD	+0.0047 $\pm$ 0.0997
GEE-Exchangeable β (95% CI; p)	+0.0038 (+0.0001 to +0.0076; $p = 0.047$)
GEE-AR(1) β (95% CI; p)	+0.0020 (-0.0013 to +0.0054; $p = 0.230$)
Cohen's d	0.047 (negligible)
% patients negative mean Δ WCS	49%

[†]WCS (per-case local PCA, sign-aligned, MinMax within case) is a distinct construct from NCCI (global PCA). WCS values are not comparable across patients. GEE analyses model Δ WCS per excursion clustered by patient.

Table 2. Surrogate excursion definition sensitivity analysis.

Definition	Cases (%)	Events	Mean CI)	Δ WCS (95%	Cohen's d	p
MAP$\geq$20, BIS<60	2,031 (98.1%)	73,001	+0.0038 +0.0076]	[+0.0001,	+0.047	0.047
MAP $\geq$ 25, BIS<60	2,013 (97.2%)	55,699	−0.0001 +0.0041]	[−0.0043,	+0.008	0.976
MAP $\geq$ 30, BIS<60	1,990 (96.1%)	45,502	−0.0040 +0.0006]	[−0.0086,	−0.026	0.089
MAP $\geq$ 20, BIS<55	2,026 (97.9%)	69,616	+0.0033 +0.0072]	[−0.0005,	+0.042	0.085
MAP $\geq$ 30, BIS<55	1,979 (95.6%)	43,539	−0.0043 +0.0004]	[−0.0089,	−0.026	0.072

GEE-Exchangeable on Δ WCS, clustered by patient. None of the stricter definitions reaches a clinically meaningful effect magnitude ($|d| \leq 0.05$). MAP $\geq$ 25 and MAP $\geq$ 30 thresholds produce near-zero or negative β values. Case rates >95% across all definitions confirm the non-specificity of the surrogate endpoint.

Patient selection flowchart

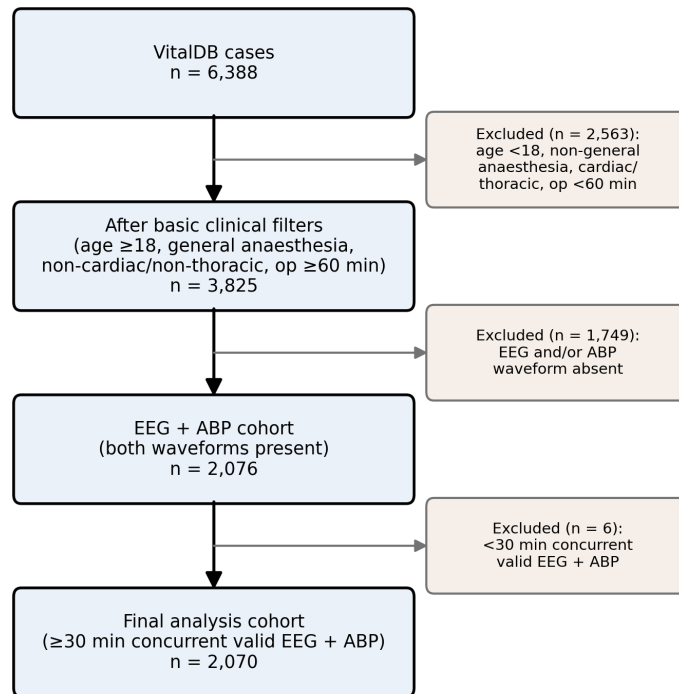


Figure 1. Patient selection flowchart. Exclusion steps from 6,388 VitalDB cases to the 2,070 cases in the coupling analysis and 2,031 cases contributing to the WCS excursion analysis.

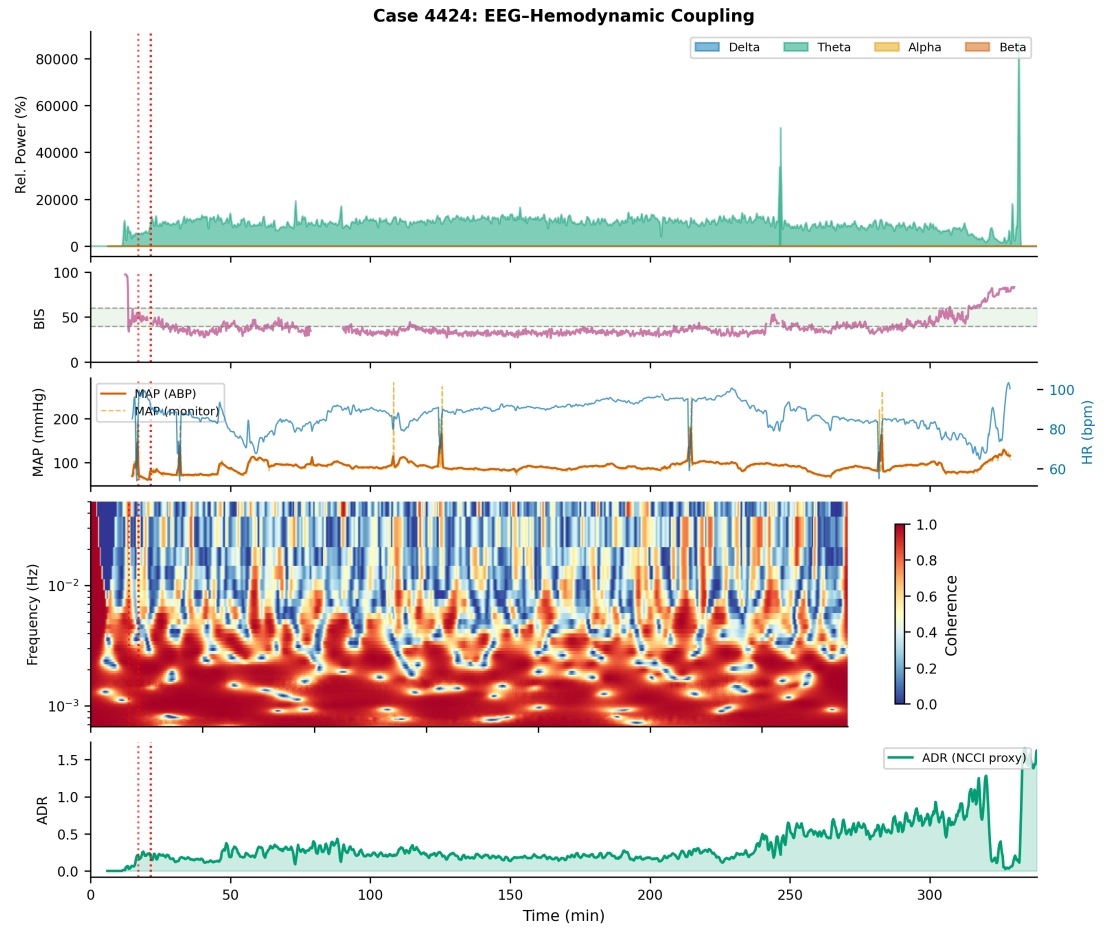


Figure 2. Representative intraoperative time series for a single case. **(A)** EEG delta and alpha band powers. **(B)** BIS value. **(C)** MAP and HR. **(D)** Cross-wavelet coherence heatmap (frequency $\times$ time). **(E)** Alpha/Delta Ratio (ADR). Time in minutes from anaesthesia start.

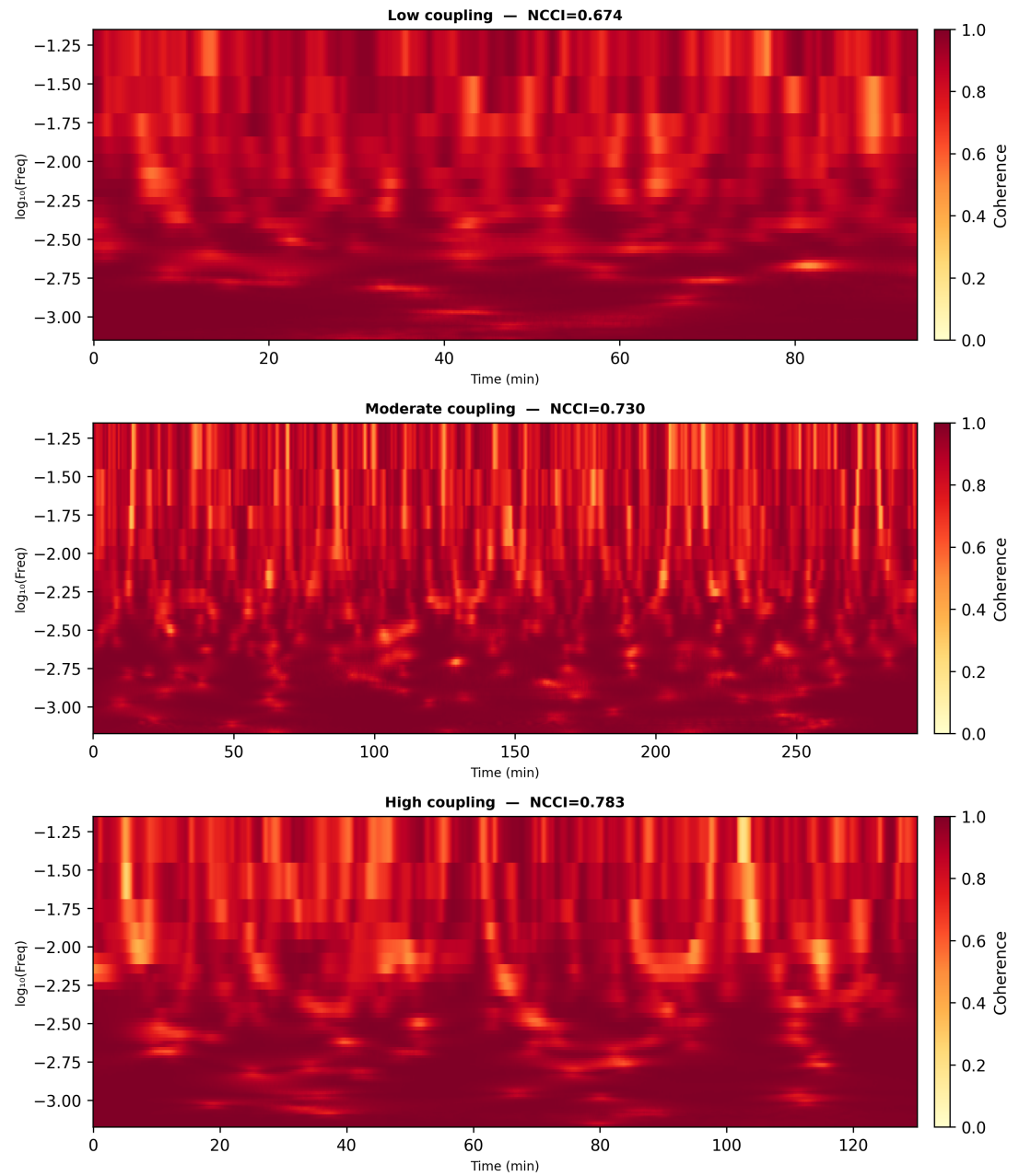


Figure 3. Wavelet coherence heatmaps for three representative cases, from the lowest (top; NCCI = 0.674), middle (centre; NCCI = 0.730), and highest (bottom; NCCI = 0.783) NCCI tertile. Colour: coherence 0–1. High low-frequency coherence (<0.01 Hz) is visible in all cases, consistent with shared pharmacodynamic slow drift.

Neurocardiac Coupling Metrics Across Cohort

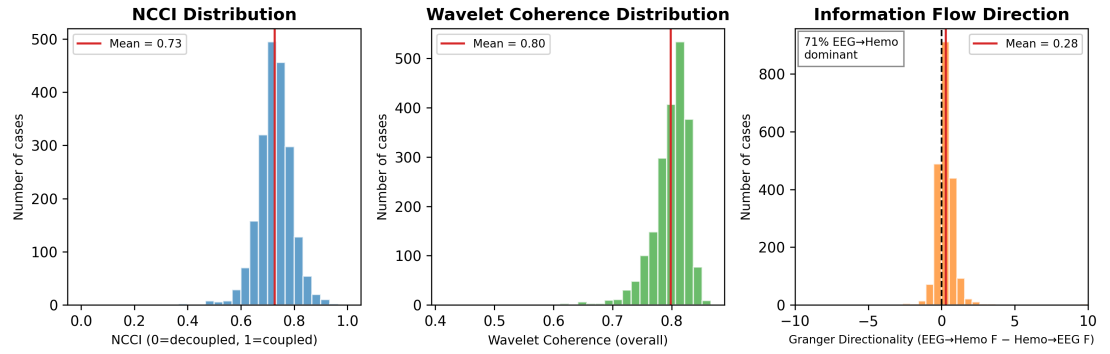


Figure 4. Distribution of case-level coupling metrics ($n = 2,070$ cases). **(A)** NCCI (composite summary score, global PCA). **(B)** WC_{overall} . **(C)** GC directionality (clipped to ± 10 for display; positive = EEG preceding MAP). Vertical lines: mean ± 1 SD.

Window Coupling Score (WCS) at Surrogate Haemodynamic Excursions ($n = 2,031$ patients, 73,001 events)

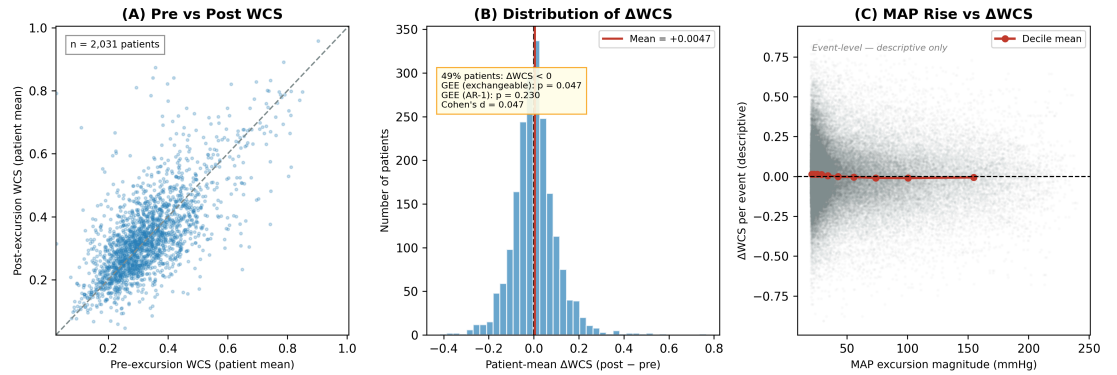


Figure 5. Window Coupling Score (WCS) at surrogate haemodynamic excursions ($n = 2,031$ patients, 73,001 events). **(A)** Paired scatter: per-patient mean pre- vs post-excursion WCS; identity line in grey. **(B)** Distribution of per-patient mean ΔWCS ; key statistical results annotated. **(C)** MAP rise magnitude vs mean ΔWCS (decile means; event-level, descriptive only). WCS (per-case local PCA) is not on the same scale as NCCI (global PCA).